

Token Averaging: Optimal Compute Allocation

1 Chinchilla Loss

$$L(N, D) = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + E \quad (1)$$

Given compute budget C and the approximation $C = 6ND$, substitute $D = C/6N$:

$$L(N) = \frac{A}{N^\alpha} + \frac{B(6N)^\beta}{C^\beta} + E \quad (2)$$

Minimize over N , set $dL/dN = 0$:

$$\frac{\alpha A}{N^{\alpha+1}} = \frac{B\beta 6^\beta N^{\beta-1}}{C^\beta} \quad (3)$$

$$N_{\text{opt}}^{\alpha+\beta} = \frac{\alpha A}{B\beta 6^\beta} \cdot C^\beta \quad (4)$$

$$\boxed{N_{\text{opt}} \propto C^{\beta/(\alpha+\beta)}} \quad (5)$$

Since $D = C/6N$:

$$\boxed{D_{\text{opt}} \propto C^{\alpha/(\alpha+\beta)}} \quad (6)$$

Define the two exponents (using Chinchilla values $\alpha = 0.3478$, $\beta = 0.2849$):

$$a = \frac{\beta}{\alpha + \beta} \approx 0.45 \quad b = \frac{\alpha}{\alpha + \beta} \approx 0.55 \quad (7)$$

2 The Compute Ratio f

FLOPs per token for a transformer with hidden dimension d and context length L :

$$\text{FLOPs/token} = N_{\text{layers}} \times L \times (24d^2 + 4dL) \quad (8)$$

where $24d^2$ is the FFN cost and $4dL$ is the attention cost per token.

Define the attention-to-FFN ratio:

$$r = \frac{4dL}{24d^2} = \frac{L}{6d} \quad (9)$$

For the **standard model at context kL** vs **base model at context L** , with the same raw token count D_{raw} :

$$f = \frac{C_{\text{std}}}{C_{\text{base}}} = \frac{24d^2 + 4d(kL)}{24d^2 + 4dL} = \frac{1 + kr}{1 + r} \quad (10)$$

$$\boxed{f = \frac{1 + kr}{1 + r}, \quad r = \frac{L}{6d}} \quad (11)$$

3 Reinvest the Saved Compute

The averaging model can use the full standard-model budget $C_{\text{std}} = f \cdot C_{\text{base}}$.

The transformer only sees $D_{\text{transformer}} = D_{\text{raw}}/k$ tokens (each group of k raw tokens is averaged into one input). So Chinchilla’s constraint becomes:

$$C = 6N \cdot D_{\text{transformer}} \quad (12)$$

This is the same Chinchilla problem, just at budget $f \cdot C_{\text{base}}$ instead of C_{base} . Apply the optimal scaling:

$$N_{\text{new}} = N_{\text{base}} \cdot f^a = N \cdot \left(\frac{1 + kr}{1 + r} \right)^{0.45} \quad (13)$$

$$D_{\text{transformer,new}} = D_{\text{base}} \cdot f^b = D_{\text{base}} \cdot \left(\frac{1 + kr}{1 + r} \right)^{0.55} \quad (14)$$

Convert back to raw tokens ($D_{\text{raw}} = k \cdot D_{\text{transformer}}$):

$$D_{\text{new}} = k \cdot D_{\text{base}} \cdot \left(\frac{1 + kr}{1 + r} \right)^{0.55} \quad (15)$$

4 Final Result

$$\boxed{N_{\text{new}} = N \cdot \left(\frac{1 + kr}{1 + r} \right)^{0.45}} \quad (16)$$

$$\boxed{D_{\text{new}} = k \cdot D_{\text{base}} \cdot \left(\frac{1 + kr}{1 + r} \right)^{0.55}} \quad (17)$$

where $r = L/6d$ is the attention-to-FFN ratio of the base model.

The $k \times$ in D_{new} is not a Chinchilla term—it is purely the conversion from transformer steps back to raw tokens. The $[(1 + kr)/(1 + r)]^{0.55}$ is the Chinchilla data scaling on top of that.